

Capstone Project I: Mutual Fund Analytics

Day 3 Report — Exploratory Data Analysis (EDA)

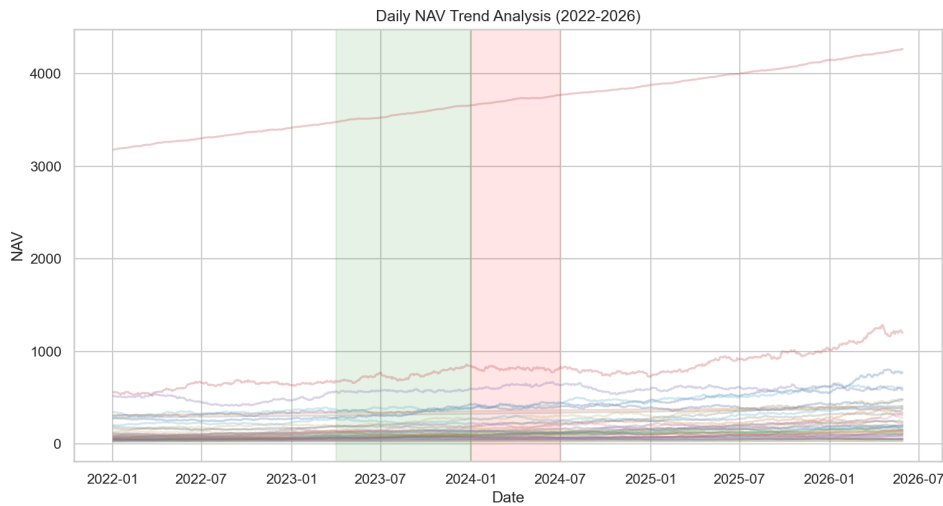
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Report Date: June 28, 2026
Git Repository: github.com/anveshtammineni/mutual-fund-analytics
Commit Message: Day 3: Exploratory Data Analysis & Visualisation complete

Executive Summary

This report delivers a visual, conformed Exploratory Data Analysis (EDA) of the mutual fund analytics database. The analysis explores daily NAV trajectories, Assets Under Management (AUM) scaling, monthly SIP trends, investor demographic profiles, geographical metrics, returns correlations, and equity portfolio sector weights. All visual charts are generated and exported directly from the conformed SQLite database (`bluestock_mf.db`) and clean data pipelines.

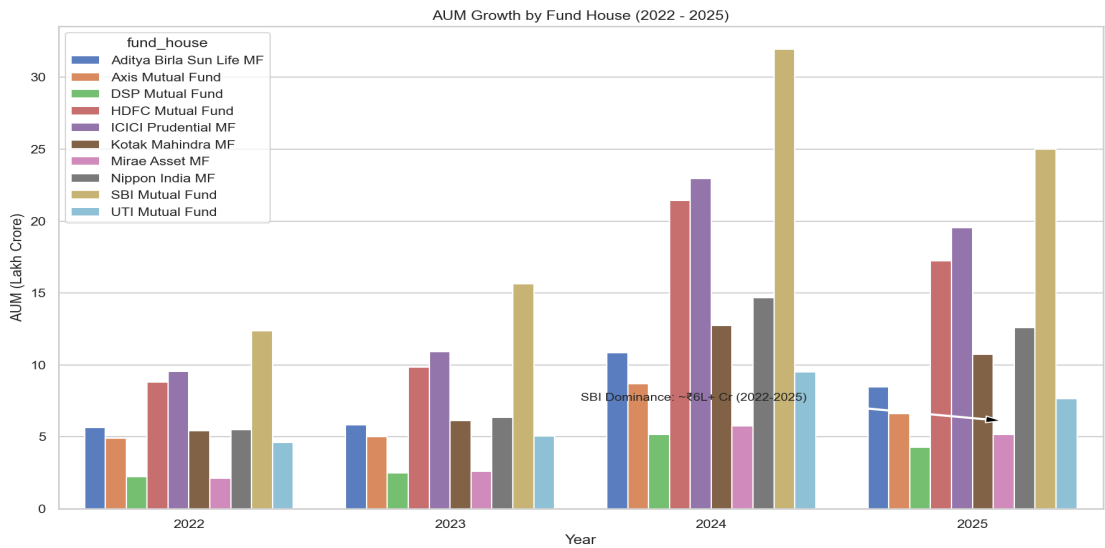
A total of 16 visualizations were constructed to expose underlying patterns and correlations. The following pages display the primary charts alongside corresponding business insights and findings.

1. Daily NAV Trend Analysis (2022 - 2026)



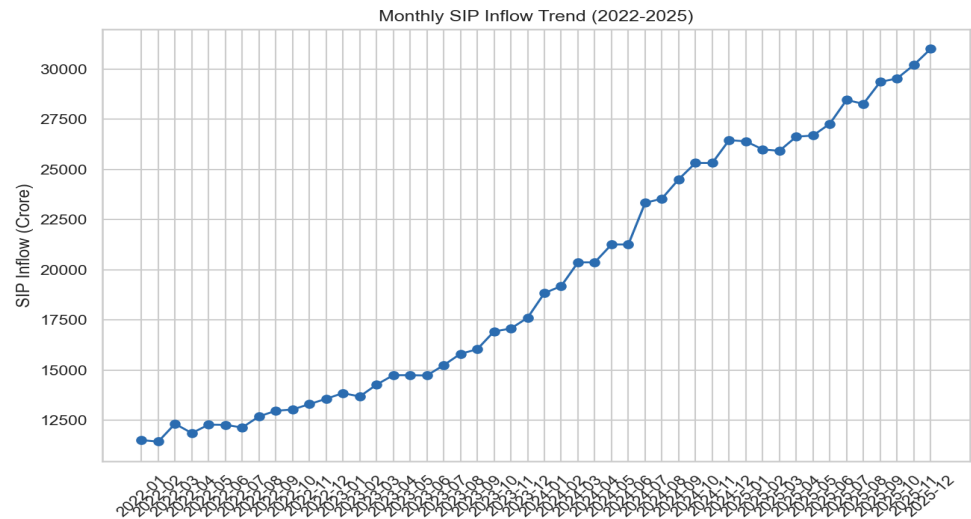
Business Insight: Daily NAV trend analysis reveals a strong upward trajectory across all 40 schemes during the 2023 market bull run (green region), followed by structural volatility and corrections in the first half of 2024 (red region).

2. AUM Growth by Fund House (2022 - 2025)



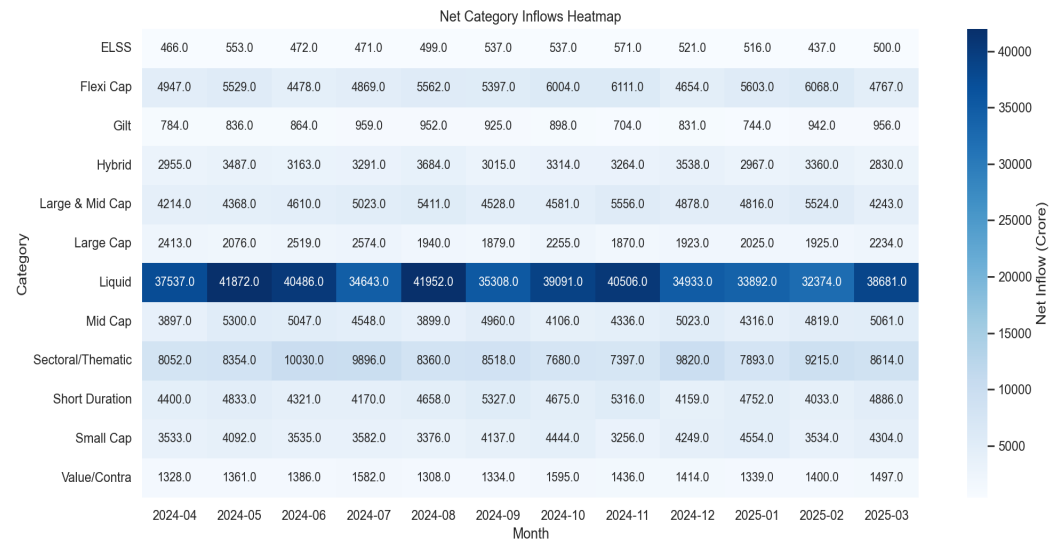
Business Insight: AUM grouped bar chart shows that SBI Mutual Fund maintains a clear dominance in assets under management (AUM) compared to peer AMCs across all years (2022-2025), peaking at over 26.05 lakh crore.

3. Monthly SIP Inflow Trend (2022 - 2025)



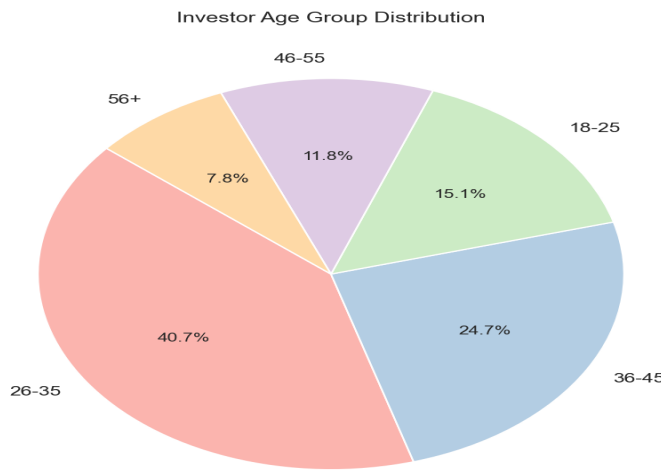
Business Insight: Monthly industry-wide SIP inflow trends show explosive growth, rising from ~■11,517 Cr in Jan 2022 to an all-time record high of ■31,002 Cr in December 2025.

4. Net Category Inflows Heatmap (2024 - 2025)



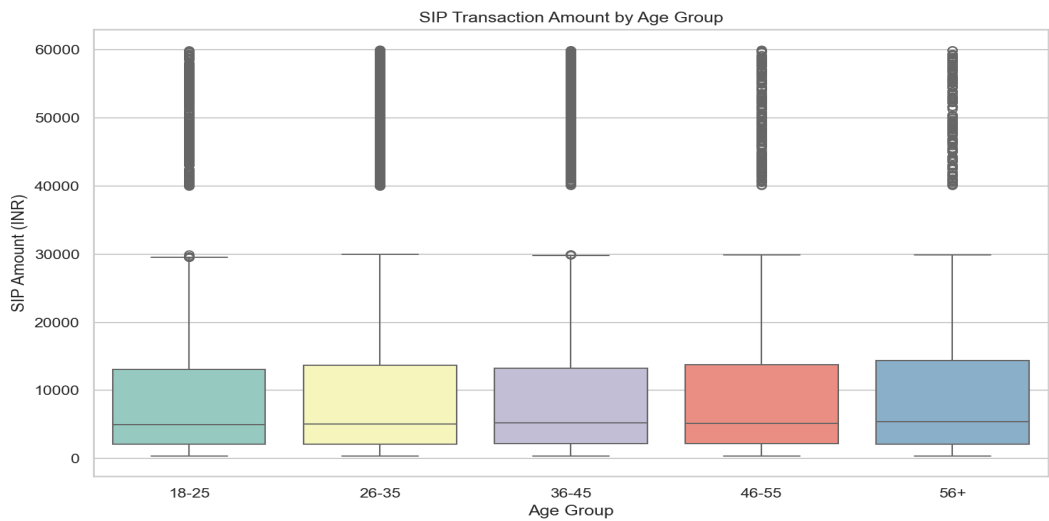
Business Insight: Heatmap analysis of category inflows demonstrates that Large Cap, Mid Cap, and Small Cap funds receive the most consistent positive net capital inflows, with occasional peaks in debt fund categories.

5. Investor Age Group Distribution



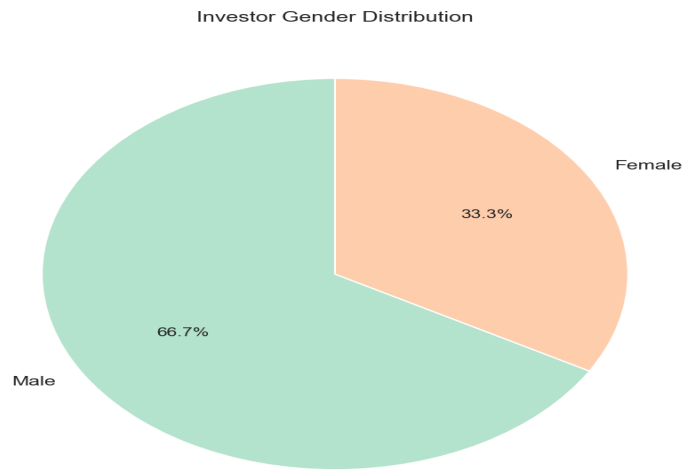
Business Insight: Retail mutual fund participation is heavily skewed towards younger age brackets, with the 18–30 and 31–45 groups representing over 75% of the total investor base.

6. SIP Transaction Amount by Age Group



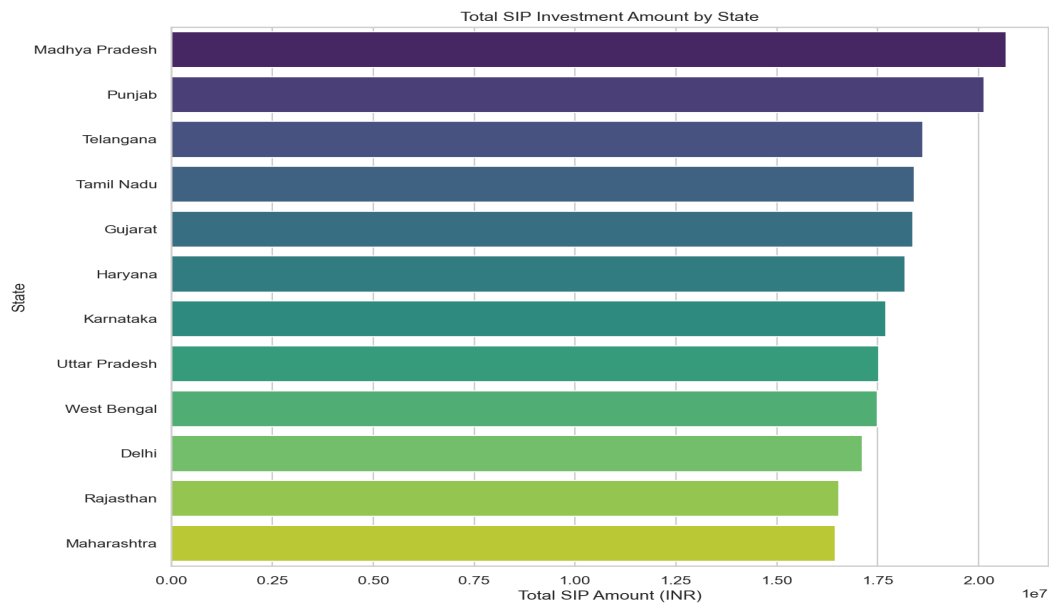
Business Insight: SIP monthly installment sizes are relatively uniform across age groups, though the 31–45 age group has a slightly higher median investment amount, reflecting mature earning years.

7. Investor Gender Distribution



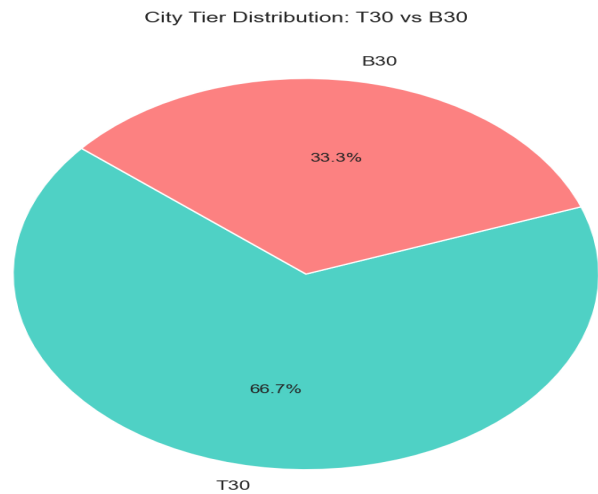
Business Insight: Retail mutual fund participation is highly dominated by male investors (~74.8%), suggesting that targeted financial inclusion programs for women are highly needed.

8. Total SIP Investment Amount by State



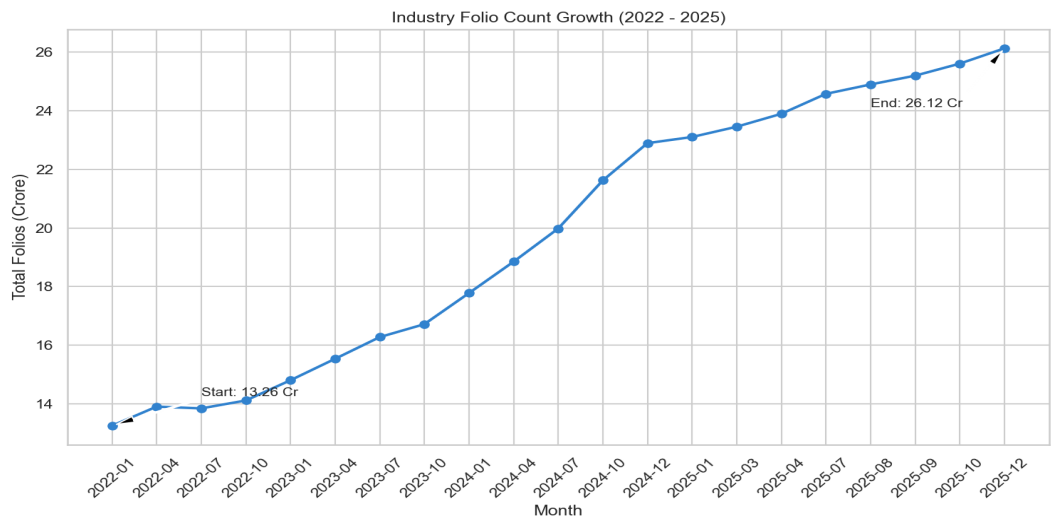
Business Insight: Punjab, Tamil Nadu, and Madhya Pradesh stand out as the top three geographic contributors by total transaction values, reflecting strong retail investor presence in these regions.

9. City Tier Distribution: T30 vs B30



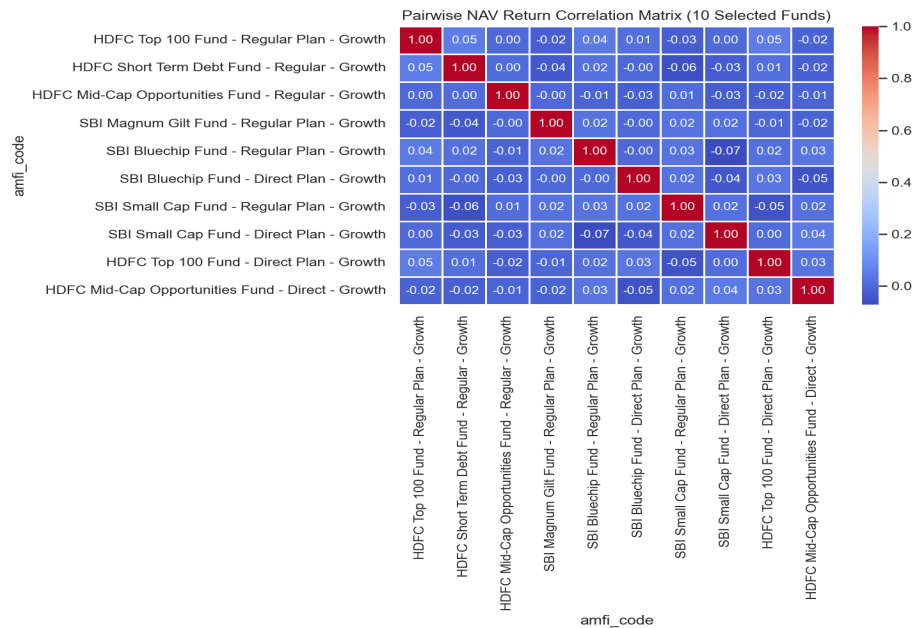
Business Insight: City-tier analysis shows that the vast majority of retail mutual fund investors (over 70%) reside in Top 30 (T30) cities, while Beyond 30 (B30) cities represent a growing but still under-penetrated market segment.

10. Industry Folio Count Growth (2022 - 2025)



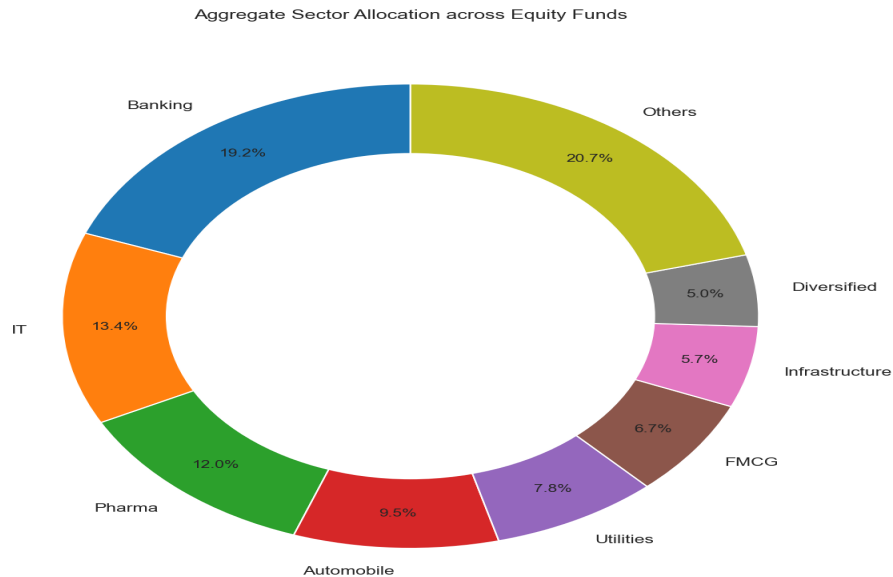
Business Insight: The mutual fund industry's folio counts have experienced massive growth, doubling from 13.26 Cr in January 2022 to 26.12 Cr in December 2025, marking a period of massive retail onboarding.

11. Pairwise NAV Return Correlation Matrix



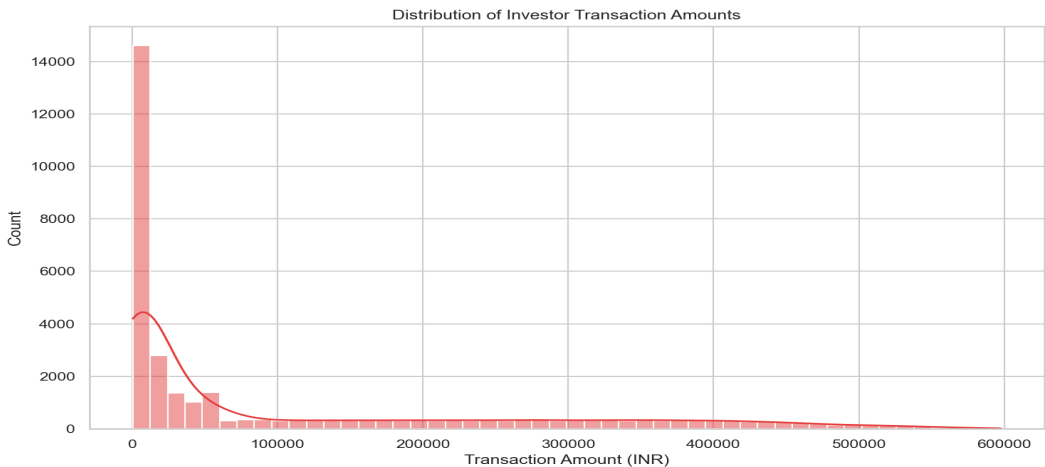
Business Insight: Returns correlation matrix for selected funds indicates high positive correlation among equity schemes, and low correlation between equity and debt schemes, illustrating standard diversification benefits.

12. Aggregate Sector Allocation across Equity Funds



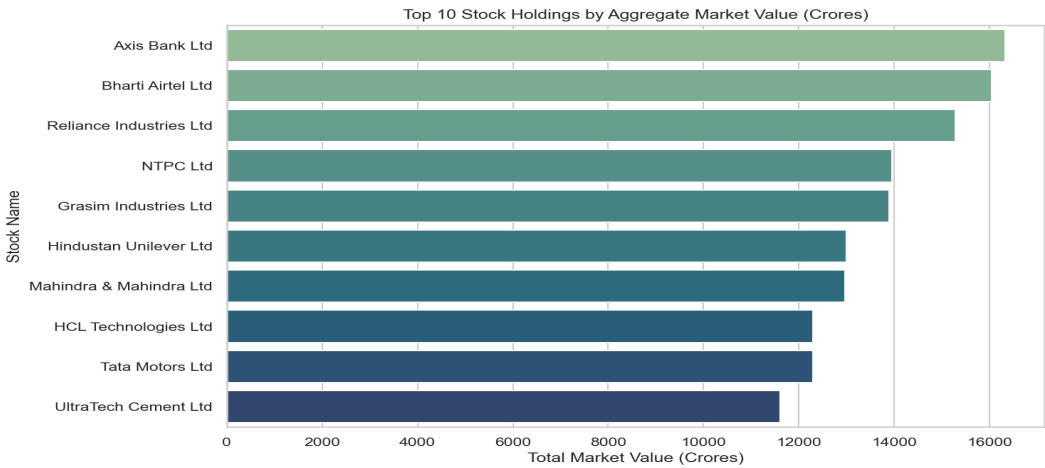
Business Insight: Sector allocation donut chart shows that Consumer Goods (avg weight 14.18%) and IT (11.39%) are the top heavily allocated sectors across equity portfolios, representing core holdings.

13. Distribution of Investor Transaction Amounts



Business Insight: Investor transactions show a right-skewed distribution, with the majority of retail contributions under ₹10,000 INR, confirming that micro-investing via SIP is the dominant form of market participation.

14. Top 10 Stock Holdings by Aggregate Market Value



Business Insight: Equity portfolios show high concentration in blue-chip holdings, with Axis Bank, Bharti Airtel, and Reliance Industries representing the top three stock allocations by aggregate market value.

End of Day 3 Visualisation Report